

Batman Attempts

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1 Mixing

We start with the derivations. I am using the notations from your notes just for consistency.

The mean we are trying to find is $\mathbb{E}_{f_0}[V(X)]$, where in previous examples we set $V(X) = X^4$. With importance sampling and mixing, we get that

$$\mathbb{E}_{f_0}[V(x)] = E_h[V(X)L(X, \theta)]$$

for

$$L(x, \theta) = \frac{f_0(x)}{pf_0(x) + (1-p)f(x, \theta)} = \frac{f_0(x)}{h(x, \theta)}$$

Then the variance is $u = \mathbb{E}_h[V(X)^2 L^2(X, \theta)] - (E_h[V(X)L(X, \theta)])^2$, and we take the partial derivative w.r.t. θ to get that

$$\begin{aligned} u_\theta &= \frac{\partial}{\partial \theta} \mathbb{E}_h[V(X)^2 L^2(X, \theta)] \\ &= \frac{\partial}{\partial \theta} \int V(x)^2 \frac{f_0(x)^2}{pf_0(x) + (1-p)f(x, \theta)} dx \\ &= \int V(x)^2 f_0(x)^2 \frac{-(1-p)f_\theta(x, \theta)}{(pf_0(x) + (1-p)f(x, \theta))^2} dx \\ &= \mathbb{E}_h[V(X)^2 L^2(X, \theta) S(X, \theta)] \end{aligned}$$

for

$$S(x, \theta) = -\frac{(1-p)f_\theta(x, \theta)}{pf_0(x) + (1-p)f(x, \theta)} = -(1-p) \frac{f_\theta(x, \theta)}{h(x, \theta)}$$

We continue the derivation and hopefully we are just getting a score function similar to the non-mixing case. From the notes,

$$f(x, \theta) = \frac{f_0(\psi(x, \theta))}{\phi_z(\psi(x, \theta), \theta)},$$

$$\partial_\theta f(x, \theta) = \partial_\theta \frac{f_0(\psi(x, \theta))}{\phi_z(\psi(x, \theta), \theta)} = \frac{\partial_\theta f_0(\psi(x, \theta))}{\phi_z(\psi(x, \theta), \theta)} - \frac{f_0(\psi(x, \theta)) \partial_\theta \phi_z(\psi(x, \theta), \theta)}{\phi_z(\psi(x, \theta), \theta)^2},$$

so we get that

$$\begin{aligned}
S(x, \theta) &= -(1-p) \frac{f_\theta(\psi(x, \theta))}{pf_0(\psi(x, \theta)) + (1-p)f(x, \theta)} \\
&= -(1-p) \frac{\partial_\theta f(\psi(x, \theta))}{pf_0(\psi(x, \theta)) + (1-p) \frac{f_0(\psi(x, \theta))}{\phi_z(\psi(x, \theta), \theta)}} \\
&= -(1-p) \frac{\partial_\theta f(\psi(x, \theta))}{\frac{pf_0(\psi(x, \theta))\phi_z(\psi(x, \theta), \theta) + (1-p)f_0(\psi(x, \theta))}{\phi_z(\psi(x, \theta), \theta)}} \\
&= -(1-p) \left(\frac{\partial_\theta f_0(\psi(x, \theta))}{f_0(z)(p\phi_z(\psi(x, \theta), \theta)) + (1-p)} - \frac{\partial_\theta \phi_z(\psi(x, \theta), \theta)}{\phi_z(\psi(x, \theta), \theta)(p\phi_z(\psi(x, \theta), \theta)) + (1-p)} \right) \\
&= \frac{-(1-p)}{p\phi_z(\psi(x, \theta), \theta) + (1-p)} \left(\frac{\partial_\theta f_0(\psi(x, \theta))}{f_0(\psi(x, \theta))} - \frac{\partial_\theta \phi_z(\psi(x, \theta), \theta)}{\phi_z(\psi(x, \theta), \theta)} \right) \\
&= \frac{(1-p)}{p\phi_z(\psi(x, \theta), \theta) + (1-p)} S_0(x, \theta) \\
&=
\end{aligned}$$

for $S_0(x, \theta) = -\left(\frac{\partial_\theta f_0(\psi(x, \theta))}{f_0(\psi(x, \theta))} - \frac{\partial_\theta \phi_z(\psi(x, \theta), \theta)}{\phi_z(\psi(x, \theta), \theta)}\right) = -(S_I(x, \theta) + S_{II}(x, \theta))$ in the notes. We can check the validity of the score function. When $p = 1$, the mixing is the same as the original sampling so $S(x, \theta) = 0$ since the variance isn't dependent on θ at all. When $p = 0$, the mixing is the same as solely using the neural net, and $S(x, \theta) = S_0(x, \theta)$, which is correct.

2 Implementation

I used $p = 0.1$ for my mixing so the neural net has the major effect. The results are satisfactory since they didn't change so much before and after mixing, which means that my distribution after neural net isn't getting very small. I can show that this is indeed the results after mixing because the best early stop is at iteration 194, while it was around 176. The new early stop parameter is $\theta = [1.45, 0.19, 0.65, 0.1]$, which is slightly different from what I got ($\theta = [1.42, 0.18, 0.64, 0.1]$).

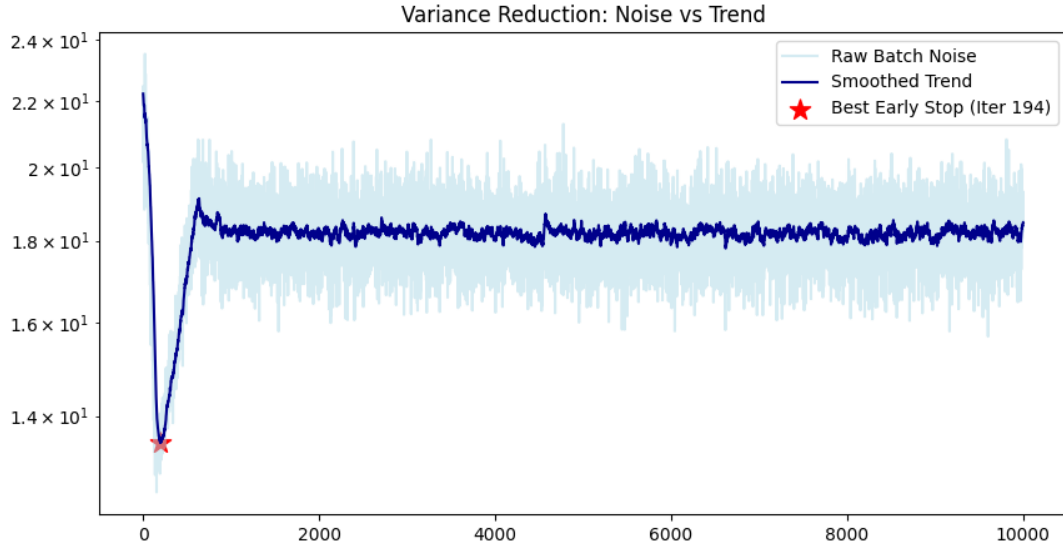


Figure 1: The variance of the neural net after early stopping and mixing.

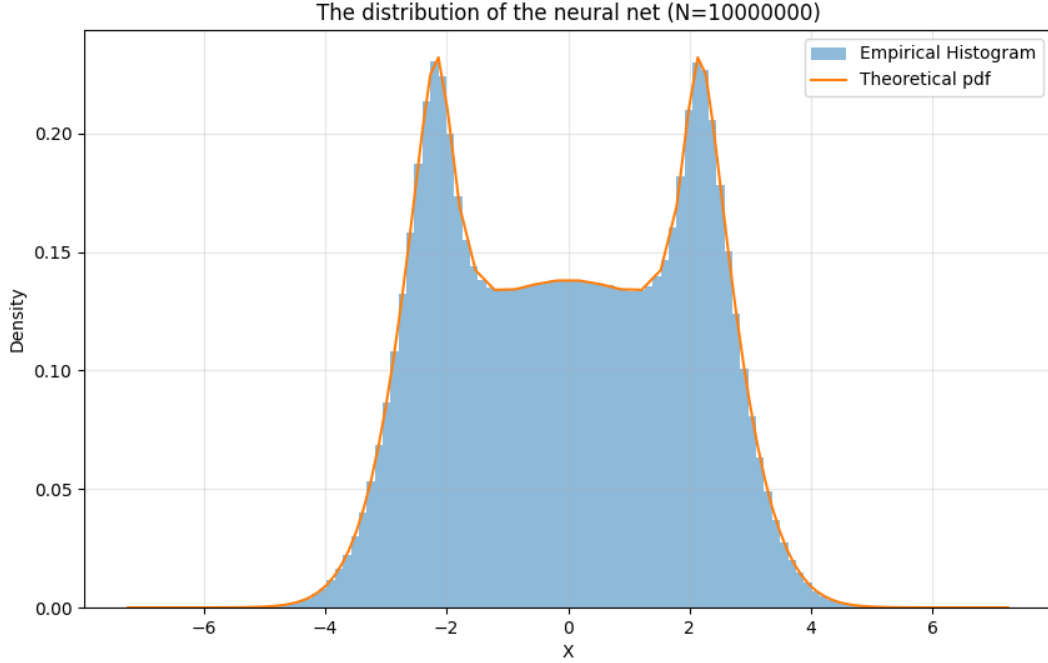


Figure 2: The distribution of the neural net after early stopping and mixing at $\theta = [1.45, 0.19, 0.65, 0.1]$.

Now instead of $p = 0.1$, I tried $p = 0.5$. I actually got almost identical results, which I cannot tell whether it's good or bad at this moment. I think it's good showing that the optimizer is robust, but $L(x, \theta)$ still changes value for different p but that's not reflected in the results.

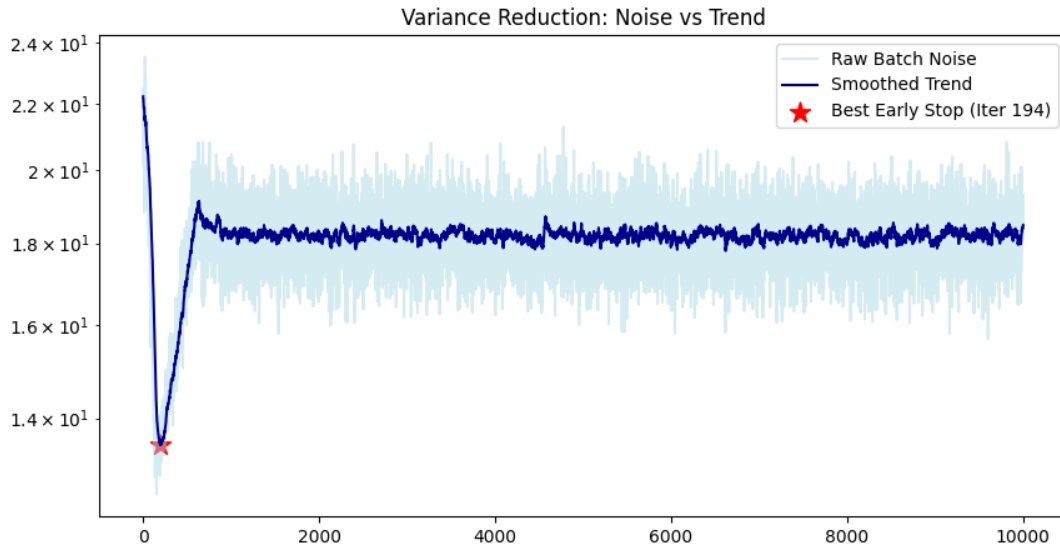


Figure 3: The variance of the neural net after early stopping and mixing.

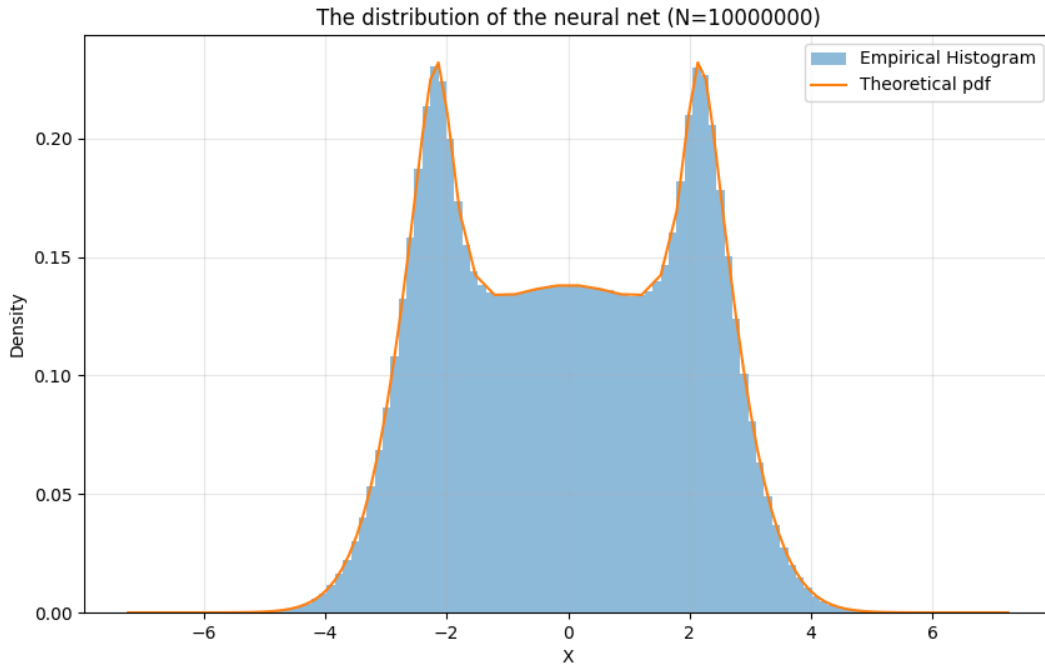


Figure 4: The distribution of the neural net after early stopping and mixing at $\theta = [1.45, 0.19, 0.65, 0.1]$.

3 Further Exploration

I did $V(X) = X^6$ out of curiosity. The variance is making me think that there's honestly something wrong with my optimizer. However, the early stopping result is actually not bad, and you can indeed see the variance dropping at the beginning.

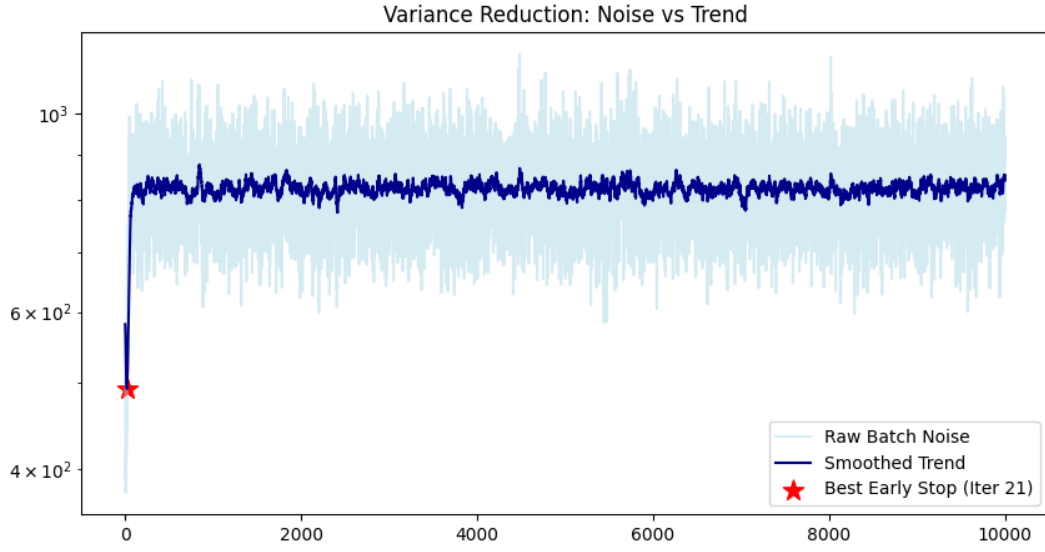


Figure 5: The variance of the neural net after early stopping and mixing.

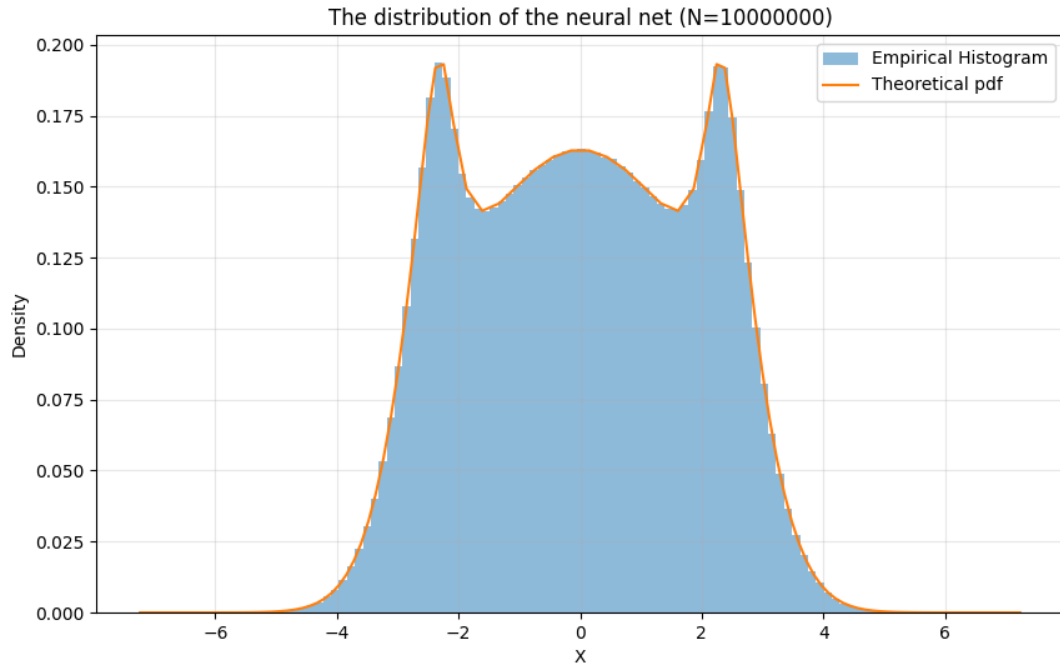


Figure 6: The distribution of the neural net after early stopping and mixing at $\theta = [1.22, 0.14, 0.86, 0.1]$.